MIL-C-7515E
23 March 1977
SUPERSEDING
MIL-C-7515D
29 June 1967
(see 6.6)

MILITARY SPECIFICATION

CORD, NYLON, CORELESS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers braided nylon coreless cord.

1.2 Classification. The cord shall be of the type specified (see table I and 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock.

MILITARY

MIL-C-3131 - Cordage; Preparation for Delivery of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4020

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Government and contractor purchases. The requirements specified in 3.7 and 3.8 apply to cord purchased directly by the Government. All other requirements apply to cord purchased by a contractor as a component for an end item and to cord purchased directly by the Government.

3.2 Standard sample. The cord shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Material. The nylon yarn used in the fabrication of the cord shall be bright, high tenacity, heat and light resistant polyamide (see 6.4) prepared from hexamethylene diamine and adipic acid or its derivatives, and shall have a melting point of 250° ± 6°C when tested as specified in 4.2.1. The nylon yarn or cord shall not be subjected to any bleaching process at any stage of processing.

* 3.4 Construction and physical requirements. The finished braided cord shall conform to the requirements specified herein and in table I, when tested as specified in 4.2.1 and 4.2.5. The cord shall be braided on braiding machines having the number of carriers specified in table I for the applicable type, and the machines shall be regulated to give a two over and two under conventional stitch, i.e., each carrier alternating with the next carrier in direction and each strand passing alternately over and under two of the opposite strands. The cord after braiding shall be heat set in continuous lengths and shall not show more than a 3 percent residual shrinkage when tested as specified in 4.2.5.

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* 3.4.1 Twist. The nylon yarns shall have the following twist:

Types Ia, IIIa, XI, XIII, XIIIa and XIV	- 1/2 to 1 turn per inch.
Types I, II, IIa, III, IV, V, VI, VII, VIII, IX, X, XV, XVI, XVII, XVIII, XIX, XX, XXI and XXII.	- Minimum of 2-1/2 turns of "Z" twist per inch.
Types XII and XIIa	- Minimum of 1 turn of "Z" twist per inch.

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TABLE I. Physical requirements

Type cord	Breaking strength, pounds (min)	No. of carriers	Ends per carrier	Total ends	Basic yarn denier	Yarn ply a/	Picks per inch	Length per pound, feet (min) b/	Elongation, percent (min)
I	400	16	3	48	210	3	11-13	330	20
Ia	400	16	2	32	840	1	13-13.5	330	20
II	550	16	3	48	840	1	10-12	255	20
IIa	c/	20	2	40	840	1	14-16	300	20
III	750	16	6	96	210	3	8.5-10	150	20
IIIa	800	24	3	72	840	1	10-12	175	20
IV	1000	16	6	96	840	1	7-9	120	20
V	1500	16	9	144	840	1	6.5-8	90	20
VI	2000	16	12	192	840	1	4.5-6.5	60	20
VII	2500	16	14	224	840	1	4.5-6	45	20
VIII	3000	24	12	288	840	1	5.5-7.5	36	20
IX	4000	24	4	96	840	4	5.5-7	27	20
X	5000	24	5	120	840	4	4.5-6	22	20
XI	300	16	7	112	210	1	14-15.5	480	20
XII	10000	36	16	576	210	7	4-5	12	20
XIIa	10000	32	18	576	210	7	4-5	12	20
XIII	50	16	4	64	70	1	22.5-26	2700	20
XIIIa	50	16	1	16	210	1	17-19	3000	20
XIV	100	16	3	48	210	1	21-24	1200	20
XV	200	16	d/	24	210	3	13-16	675	20
XVI	1250	16	7	112	840	1	6.5-8	105	20
XVII	1750	16	10	160	840	1	5.5-7	75	20
XVIII	2250	16	13	208	840	1	5-6	51	20
XIX	3500	24	14	336	840	1	5.5-7	30	20
XX	4500	24	e/	108	840	4	5.5-6.5	24	20
XXI	5500	24	f/	132	840	4	4.5-5.5	18	20
XXII	6000	24	6	144	840	4	4.5-5.5	15	20

a/ Ten or less linear yards of ply yarns shall not contain any knots.

b/ A minus tolerance of 10 percent in length will be allowed for dyed cord.

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c/ Breaking strength for type IIa cord shall be 550 ± 20 pounds.

d/ Eight carriers of one end each, eight carriers of two ends each, alternately.

e/ Twelve carriers of four ends each, twelve carriers of five ends each, alternately.

f/ Twelve carriers of five ends each, twelve carriers of six ends each, alternatively.

* 3.4.2 Carrier run-off. In the manufacture of the cord, no more than one carrier shall be allowed to be run off per length of cord specified in table II for the applicable type. When carrier ends run off, they shall be spliced a minimum distance of 5 inches to a maximum of 10 inches in length. When splicing the outgoing carrier ends shall be joined with the new ends and hand braided for four to five picks and the tail of the splice shall be run into the center of the braided cord. The approximate center of each splice of type IIa cord shall be marked by inserting a bright red thread (of another fiber type) through the braid. The free ends of the marker shall extend approximately 3 inches on either side of the cord. The marker shall not be removed until the cord is used in the manufacture of parachute suspension lines or other applications as specified (see 6.1).

TABLE II. Carrier run-off

Types	Distance between carrier run-offs feet (min) 1/
I, Ia, XI, XIII, XIIIa, XIV	100
II, III	50
IIa, IIIa, IV	40
XVI	30
V	25
VI, XVII	20
VII, VIII, XV, XVIII, XIX	10
IX, X, XXII	5
XII, XIIa, XX, XXI	3

1/ A minus tolerance of 10 percent in length will be allowed on dyed cord.

3.5 Color. Unless otherwise specified (see 6.2), the color of the cord shall be natural. When colored cord is specified, the cord shall be piece dyed to match the approved standard shade of Olive Drab 7 (see 6.3). The cord shall be piece dyed under uniform tension at all times during the dyeing process. The dyeing of cord in skeins is prohibited.

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3.5.1 Matching. The color of the dyed cord shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500 Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 Kelvin.

3.5.2 Colorfastness. The dyed cord shall show colorfastness to light and laundering equal to or better than the standard sample when tested as specified in 4.2.5. When no standard sample has been established, the dyed cord shall show good fastness to light and laundering when tested as specified in 4.2.5.

3.6 Extractable matter. The chloroform soluble material of the finished cord shall not exceed 2.0 percent when tested as specified in 4.2.5.

* 3.7 Put-up. Unless otherwise specified (see 6.2), the cord shall be furnished in continuous length 700-yard spools (reels), except type I, IV, VI, and XI lengths shall be as specified by the procuring activity (see 6.2), without knots or splices and wound so that each turn and layer is free from entanglement. A plus tolerance of 10 percent shall be allowed on the length specified. The ends of all cords shall be taped, served, or heat sealed to prevent fraying.

* 3.8 Identification. Each spool (reel) of cord shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, grade 1SSU or 1SCSU, size 4 or 5 of UU-T-81. When labels are used, the label shall be attached in such a manner as to remain in place and be clearly legible until all cord has been removed. The ticket or label shall be legibly printed, stamped, or typed with water insoluble ink. Handwritten entries shall be prohibited. The ticket or label shall contain the following information:

Stock number
Nomenclature
Specification number
Length
Contract number and date
Date of manufacture (month and year)
Contractor's name

3.9 Workmanship. The finished cord shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling and inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, testing shall be performed for the requirements in table III and will be accepted on a contractor's certificate of compliance thereto.

TABLE III. Component tests

Characteristic	Requirement reference	Test method
Material identification	3.3	1530 FED-STD-191
Tenacity	3.3	
Heat and light resistant yarn	3.3	
Melting point	3.3	1534 FED-STD-191
Bleaching of nylon yarn or cord	3.3	
Twist	3.4.1	4054 FED-STD-191
Denier	Table I	4021 FED-STD-191
Ply	Table I	Visual

* 4.2.2 In-process inspection. Inspection shall be conducted to see that knots in plied yarns do not exceed requirements specified in table I and that accomplishment of splicing carrier run-off is in accordance with 3.4.2 and that when specified, the colored cord is in accordance with 3.5. Unless otherwise specified, a contractor's certificate of compliance thereto shall be furnished and will be accepted for the requirement.

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* 4.2.3 Examination of the end item. The end item shall be examined in accordance with the classification of defects set forth in 4.2.3.1 and 4.2.3.2 at the inspection levels and acceptable quality levels (AQLs) specified.

* 4.2.3.1 Examination of the end item for visual defects. The defects specified in table IV shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition in which case only the more serious defect shall be counted. The sample unit for the examination shall be one spool (reel). Ten percent of the length contained on each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of spools (reels). The lot shall consist of all spools (reels) offered for inspection at one time. The inspection level shall be level III. The AQL shall be 1.5 major defects, and 6.5 total defects (major and minor combined) per 100 units. There shall be no critical defects in the samples examined.

TABLE IV. Visual examination

Examine	Defect	Classification		
		Critical	Major	Minor
Appearance and workmanship	Any cut.	X		
	Abrasion, chafed area or distortion in the orientation of yarns.	X		
	Kinks, knots or unevenly braided resulting in open places, breaks in continuity of braid, or soft spots.	X		
	Floats: broken, projecting or missing ends or picks.	X		
Type	Other than specified.	X		
Color	Other than specified.		X	
Cleanness	Spot or stain - clearly noticeable 1/.			X
	Objectionable odor.			X
Identification	Omitted, incorrect, illegible, insecurely attached.			X
	Ticket or label not as specified.			X
	Handwritten entries.			X

1/ At normal inspection distance (approximately 3 feet).

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* 4.2.3.2 Examination for length and winding. The sample unit for this examination shall be one spool (reel). The lot shall consist of all spools (reels) offered for inspection at one time. For lots consisting of over 500 units, the inspection level shall be S-3 and the AQL shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number 1. Defects shall be as listed in 4.2.3.2.1 and 4.2.3.2.2.

* 4.2.3.2.1 Length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

(a) The length is less than specified or more than 10 percent in excess of the length specified.
(b) Length of cord on spool (reel) more than 2 yards less than marked on ticket or label.

4.2.3.2.2 Winding. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

(a) Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding, or otherwise affecting free unhampered unwinding of cord.
(b) Put-up not as specified.
(c) Any end not taped, served, or heat sealed.
(d) Any knot or splice.
(e) Cord not in a continuous length.

* 4.2.4 Inspection of preparation for delivery requirements. Packaging, packing and marking shall be inspected in accordance with quality assurance provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

* 4.2.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table VI shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table V. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for testing shall be one spool (reel) of cord. The lot shall be unacceptable if one or more units fail to meet any requirement. The lot size shall be expressed in units of spools (reels). When the data in the Number of Determinations per sample unit and Results reported as columns are not specified in table VI, they shall be as required by the referenced test method.

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TABLE V. Sampling for tests

No. of spools (reels) in lot	Sample size
15 and under	2
16 up to and including 40	3
41 up to and including 110	5
111 up to and including 300	7
301 up to and including 500	10
501 and over	15

TABLE VI. Test methods 1/

Characteristic	Requirement reference	Test method	No. of deter. per sample unit	Results reported as
Breaking strength and elongation	Table I	6016	-	-
Number of carriers	Table I	Visual	3	Pass or fail
Ends per carrier	Table I	Visual	3	Pass or fail
Total ends	Table I	Visual	3	Pass or fail
Picks per inch	Table I	6001	-	-
Length per pound	Table I	6004	-	-
Shrinkage	3.4	6010	-	-
Dyeing of cord in skeins	3.5		-	-
Colorfastness to:				
Light	3.5.2	5660 2/	-	-
Laundering	3.5.2	5614	-	-
Extractable matter	3.6	2611	-	-

1/ Test to determine compliance with specification requirements including quantity of delivery may be made under prevailing atmospheric conditions except in settlement of dispute in which case the tests shall be made upon material which has reached equilibrium under Standard Conditions as defined in FED-STD-191.

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2/ The specimen shall be 2 feet or more of the cord wrapped around a 6-inch square of cardboard which, after the required exposure period, shall be removed from the machines and placed in the dark a minimum of 1 hour before evaluation.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A and C. The cord, put up on spools (reels), shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B, and C as specified (see 6.2).

5.2.1 Levels A, B and C. The cord shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

6. NOTES

* 6.1 Intended use. The cord is intended for use in cargo type parachutes. Type XI cord is intended for use as tow cables. Type Ia and IIIa cord is intended for low cost parachutes. Type IIa cord is intended for use as a weak link in a personnel parachute line.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2 and table I).
(c) When color Olive Drab 7 is required (see 3.5).
(d) Put-up length required when type I, IV, VI or XI is specified (see 3.7).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to the standard samples, address the procuring office issuing the invitation for bids.

6.4 Yarn. Ultraviolet resistant yarn of DuPont type 702 and 330, and Monsanto Type A02 are acceptable (see 3.3).

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Supersession data. This specification includes the requirements of Purchase Description LP/P DES 4-70 dated 10 February 1970.

Custodians:

Army - GL
Navy - AS
Air Force - 99

Review activities:

Army - MD, MU
Navy - SA

User activities:

Army - AV, ME, WC
Navy - OS, MC

Preparing activity:

Army - GL

Project No. 4020-0219

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